DATABASE MANAGEMENT LAB Report

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

to

K.R Mangalam University

Submitted by:

RADHIKA

(2401010094)

Submitted to

Ms. Mansi Kajal

(Assistant Professor)

Department of Computer Science and Engineering School of Engineering and Technology K.R Mangalam University, Gurugram- 122001, India SESSION – 2025-2026

Objectives

- **To understand the use of COMMIT and ROLLBACK commands in transaction control.**
- **To use ORDER BY and GROUP BY clauses in SQL queries.**
- **To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.**
- **To understand and implement JOIN operations in SQL.**

DDL Questions (Data Definition Language)

1. **Create the TEACHER table with the following attributes: (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).**

```sql
CREATE DATABASE Radhikalab_assignment_3;
USE Radhikalab_assignment_3;
CREATE TABLE TEACHER (
    Name VARCHAR(50),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary DECIMAL(10,2)
);
```

2. **Create the SALES table with the following attributes: (Sales_No, Salesname, Branch, Salesamount, DOB).**

```sql
CREATE TABLE SALES (
    Sales_No INT PRIMARY KEY,
    Salesname VARCHAR(50),
    Branch VARCHAR(50),
    Salesamount DECIMAL(10,2),
    DOB DATE
);
```

3.Create the DEPARTMENTS table with the following attributes: (dept_no, dept_name, dept_location).

```
CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
```

4.Create the EMPLOYEES table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

```
CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

DML Questions (Data Manipulation Language)

<u>From TEACHER Table</u>

1. **Insert five records into the TEACHER table.**

```
INSERT INTO TEACHER VALUES
('Radhika', 1, '2018-06-15', 'Mathematics', 'Delhi', 50000),
('Nikita', 2, '2019-07-20', 'Commerce', 'Mumbai', 45000),
('mansi', 1, '2017-03-12', 'Mathematics', 'Delhi', 55000),
('Simi', 3, '2020-01-10', 'Physics', 'Chennai', 40000),
('Ashi', 2, '2016-09-05', 'Commerce', 'Mumbai', 48000);
SELECT* FROM TEACHER ;
```

<u>OUTPUT :</u>

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Radhika	1	2018-06-15	Mathematics	Delhi	50000.00
Nikita	2	2019-07-20	Commerce	Mumbai	45000.00
mansi	1	2017-03-12	Mathematics	Delhi	55000.00

2. **Give an increment of 25% salary to teachers working in the Mathematics department.**

```
SET SQL_SAFE_UPDATES = 0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
SELECT* FROM TEACHER;
```

OUTPUT :

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Radhika	1	2018-06-15	Mathematics	Delhi	62500.00
Nikita	2	2019-07-20	Commerce	Mumbai	45000.00
mansi	1	2017-03-12	Mathematics	Delhi	68750.00
Simi	3	2020-01-10	Physics	Chennai	40000.00
Ashi	2	2016-09-05	Commerce	Mumbai	48000.00

3. **Give an increment of 15% salary to teachers working in the Commerce department.**

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
SELECT* FROM TEACHER;
```

OUTPUT :

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Radhika	1	2018-06-15	Mathematics	Delhi	62500.00
Nikita	2	2019-07-20	Commerce	Mumbai	51750.00
mansi	1	2017-03-12	Mathematics	Delhi	68750.00
Simi	3	2020-01-10	Physics	Chennai	40000.00
Ashi	2	2016-09-05	Commerce	Mumbai	55200.00

From SALES Table

1. **Insert five records into the SALES table.**

```
INSERT INTO SALES VALUES
(101, 'Rashi', 'North', 20000, '1998-12-21'),
(102, 'Priyanka', 'South', 30000, '1997-05-10'),
(103, 'Radhika', 'East', 25000, '1996-12-11'),
(104, 'Meenakshi', 'West', 40000, '1995-08-15'),
(105, 'Vamika', 'North', 15000, '1999-12-05');
SELECT* FROM SALES;
```

OUTPUT :

Result Grid | Filter Rows: | Edit:

Sales_No	Salesname	Branch	Salesamount	DOB
101	Rashi	North	20000.00	1998-12-21
102	Priyanka	South	30000.00	1997-05-10
103	Radhika	East	25000.00	1996-12-11
104	Meenakshi	West	40000.00	1995-08-15
105	Vamika	North	15000.00	1999-12-05
NULL	NULL	NULL	NULL	NULL

2. **Calculate the total sales amount in each branch.**

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

Result Grid | Filter

Branch	Total_Sales
North	35000.00
South	30000.00
East	25000.00
West	40000.00

3. **Calculate the average sales amount in each branch.**

```
SELECT Branch, AVG(Salesamount) AS Avg_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT:

Result Grid | Filter R

Branch	Avg_Sales
North	17500.000000
South	30000.000000
East	25000.000000
West	40000.000000

4. **Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.**

```
SELECT Salesname,
DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

OUTPUT:

Result Grid | Filter Rov

Salesname	DOB
Rashi	21-Dec-98
Radhika	11-Dec-96
Vamika	05-Dec-99

5. **Display the Sales name and DOB of salesmen in alphabetical order of the month.**

```
SELECT Salesname, DOB
```

Result Grid | Filter Rov

Salesname	DOB
Meenakshi	1995-08-15
Rashi	1998-12-21
Radhika	1996-12-11
Vamika	1999-12-05
Priyanka	1997-05-10

```
AME(DOB);
```

From JOIN Operations

1. **Display employee details and department details where the department numbers are the same in both tables.**

```
INSERT INTO DEPARTMENTS VALUES
(1, 'Mathematics', 'Delhi'),
(2, 'Commerce', 'Mumbai'),
(3, 'Physics', 'Gurugram');
SELECT* FROM DEPARTMENTS;
```

OUTPUT :

Result Grid | Filter Rows:

dept_no	dept_name	dept_location
1	Mathematics	Delhi
2	Commerce	Mumbai
3	Physics	Gurugram
NULL	NULL	NULL

```sql
INSERT INTO EMPLOYEES VALUES
(1, 'Ankit', 50000, 1),
(2, 'Bhavika', 45000, 2),
(3, 'Gaurav', 60000, 1),
(4, 'Diksha', 35000, 3),
(5, 'Ishika', 70000, 2);
SELECT* FROM EMPLOYEES;
```

OUTPUT:

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
1	Ankit	50000.00	1
2	Bhavika	45000.00	2
3	Gaurav	60000.00	1
4	Diksha	35000.00	3
5	Ishika	70000.00	2
NULL	NULL	NULL	NULL

```sql
SELECT *
FROM EMPLOYEES E
INNER JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
1	Ankit	50000.00	1	1	Mathematics	Delhi
3	Gaurav	60000.00	1	1	Mathematics	Delhi
2	Bhavika	45000.00	2	2	Commerce	Mumbai
5	Ishika	70000.00	2	2	Commerce	Mumbai
4	Diksha	35000.00	3	3	Physics	Gurugram

2. **Display the employee name and department name using a LEFT OUTER JOIN.**

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

Result Grid | Filter Rows

emp_name	dept_name
Ankit	Mathematics
Bhavika	Commerce
Gaurav	Mathematics
Diksha	Physics
Ishika	Commerce

3. **Display the employee name and department name using a RIGHT OUTER JOIN.**

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT:

Result Grid | Filter Row

emp_name	dept_name
Ankit	Mathematics
Gaurav	Mathematics
Bhavika	Commerce
Ishika	Commerce
Diksha	Physics

4. **Display the details of employees who draw a salary greater than the average salary.**

```
SELECT *
FROM EMPLOYEES
WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

OUTPUT:

Result Grid | Filter Rows:

emp_id	emp_name	emp_salary	dept_no
3	Gaurav	60000.00	1
5	Ishika	70000.00	2
NULL	NULL	NULL	NULL

TCL Questions (Transaction Control Language)

1. **Perform the ROLLBACK command on the TEACHER table.**

```
SET autocommit = 0;
UPDATE TEACHER
SET Salary = Salary + 1000;
ROLLBACK;
SELECT * FROM TEACHER;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Cor

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Radhika	1	2018-06-15	Mathematics	Delhi	63500.00
Nikita	2	2019-07-20	Commerce	Mumbai	52750.00
mansi	1	2017-03-12	Mathematics	Delhi	69750.00
Simi	3	2020-01-10	Physics	Chennai	41000.00
Ashi	2	2016-09-05	Commerce	Mumbai	56200.00

2. **Perform the COMMIT command on the TEACHER table**.

```
UPDATE TEACHER
SET Salary = Salary + 1000;
COMMIT;
SELECT* FROM TEACHER;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Conten

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Radhika	1	2018-06-15	Mathematics	Delhi	64500.00
Nikita	2	2019-07-20	Commerce	Mumbai	53750.00
mansi	1	2017-03-12	Mathematics	Delhi	70750.00
Simi	3	2020-01-10	Physics	Chennai	42000.00
Ashi	2	2016-09-05	Commerce	Mumbai	57200.00

3. **Write a query to implement a SAVEPOINT.**

```
UPDATE TEACHER SET Salary = Salary + 100;
SAVEPOINT A;
SELECT* FROM TEACHER;
UPDATE TEACHER SET Salary = Salary + 1000;
ROLLBACK TO A;
SELECT* FROM TEACHER;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Conten

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Radhika	1	2018-06-15	Mathematics	Delhi	64600.00
Nikita	2	2019-07-20	Commerce	Mumbai	53850.00
mansi	1	2017-03-12	Mathematics	Delhi	70850.00
Simi	3	2020-01-10	Physics	Chennai	42100.00
Ashi	2	2016-09-05	Commerce	Mumbai	57300.00

DCL Questions (Data Control Language)

1.Write a query to grant all privileges on the EMPLOYEES table to DEPARTMENTS.

```
CREATE USER 'departments_user'@'localhost'
IDENTIFIED BY '1234';
GRANT ALL PRIVILEGES
ON Radhikalab_assignment_3
TO 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

2.Write a query to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT SELECT, INSERT, UPDATE
ON Radhikalab_assignment_3
TO 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

3.Write a query to revoke all privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE ALL PRIVILEGES
ON Radhikalab_assignment_3
FROM 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
GRANT SELECT, INSERT, UPDATE
ON Radhikalab_assignment_3
TO 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```

4.Write a query to revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE SELECT, INSERT
ON Radhikalab_assignment_3
FROM 'departments_user'@'localhost';
SHOW GRANTS FOR 'departments_user'@'localhost';
```